Atlas Research Lab — Internal Knowledge Guide

This document is designed as a clean test corpus for a Retrieval-Augmented Generation (RAG) system. It contains distinct facts, definitions, procedures, comparisons, and numerical values so that retrieval can be tested with questions requiring different parts of the document.

1. Organization Overview

Atlas Research Lab is a fictional technology organization founded in 2021. Its main office is located in Beirut, Lebanon. The laboratory develops software for document intelligence, knowledge retrieval, workflow automation, and data analysis. Its engineering team is divided into four groups: Data Systems, Language Intelligence, Platform Engineering, and Evaluation.

2. Team Responsibilities

Team	Primary responsibility	Main output
Data Systems	Data ingestion and preprocessing	Clean datasets and pipelines
Language Intelligence	NLP, embeddings, and LLM applications	Language models and retrieval components
Platform Engineering	APIs, infrastructure, and deployment	Production services
Evaluation	Quality measurement and testing	Benchmarks and evaluation reports

3. Atlas Knowledge Retrieval System

The Atlas Knowledge Retrieval System uses a two-stage process. First, a retriever searches a vector index for passages that are semantically close to the user's question. Second, a language model receives the selected passages and generates an answer grounded in that context. The system should not invent information that is absent from the retrieved material.

The default embedding model for the current prototype is Sentence Transformer MiniLM. The vector database contains 12,480 indexed chunks. Each chunk is normally limited to approximately 500 characters, with an overlap of 100 characters between neighboring chunks.

4. Retrieval Configuration

The similarity retriever uses cosine similarity. A normal query retrieves the top 4 candidate chunks. If the highest similarity score is below 0.55, the system marks the retrieval result as weak. A score from 0.55 to 0.74 is considered moderate, while a score of 0.75 or higher is considered strong. These thresholds are evaluation conventions for the prototype and are not universal rules for every RAG system.

5. Evaluation Rules

RAG evaluation is divided into retrieval quality and generation quality. Retrieval quality asks whether the relevant information was found. Generation quality asks whether the final answer correctly uses that information. A system may have strong retrieval but poor generation if the model ignores the retrieved context. Conversely, fluent generation cannot compensate for retrieving irrelevant evidence.

For the Atlas prototype, the evaluation team records four measurements: retrieval precision, retrieval recall, answer faithfulness, and answer completeness. Each measurement is scored from 0 to 1.

6. Incident Response Procedure

1. Confirm that the API is reachable.

2. Check whether the retriever returned any chunks.

3. Inspect the similarity scores of the returned chunks.

4. Verify that the prompt contains the retrieved context.

5. Check the language-model response for unsupported claims.

6. Record the incident and the final diagnosis in the evaluation log.

7. Project Milestones

Milestone	Date	Status
Initial prototype	March 2021	Completed
Vector retrieval added	September 2022	Completed
Evaluation framework	May 2024	Completed
Agent controller	February 2025	Completed
Production pilot	October 2026	Planned

8. Security Policy

Retrieved documents may contain confidential business information. The retrieval service must therefore log access events without storing the complete contents of confidential documents in ordinary application logs. API credentials must be stored in environment variables or a dedicated secret-management system. Production access should use authentication and role-based authorization.

9. Key Facts for Testing

• Atlas Research Lab was founded in 2021.

• The main office is in Beirut, Lebanon.

• The vector index contains 12,480 chunks.

• The default retrieval count is 4 chunks.

• A similarity score of 0.75 or above is classified as strong.

• The current prototype uses Sentence Transformer MiniLM.

• The planned production pilot is scheduled for October 2026.